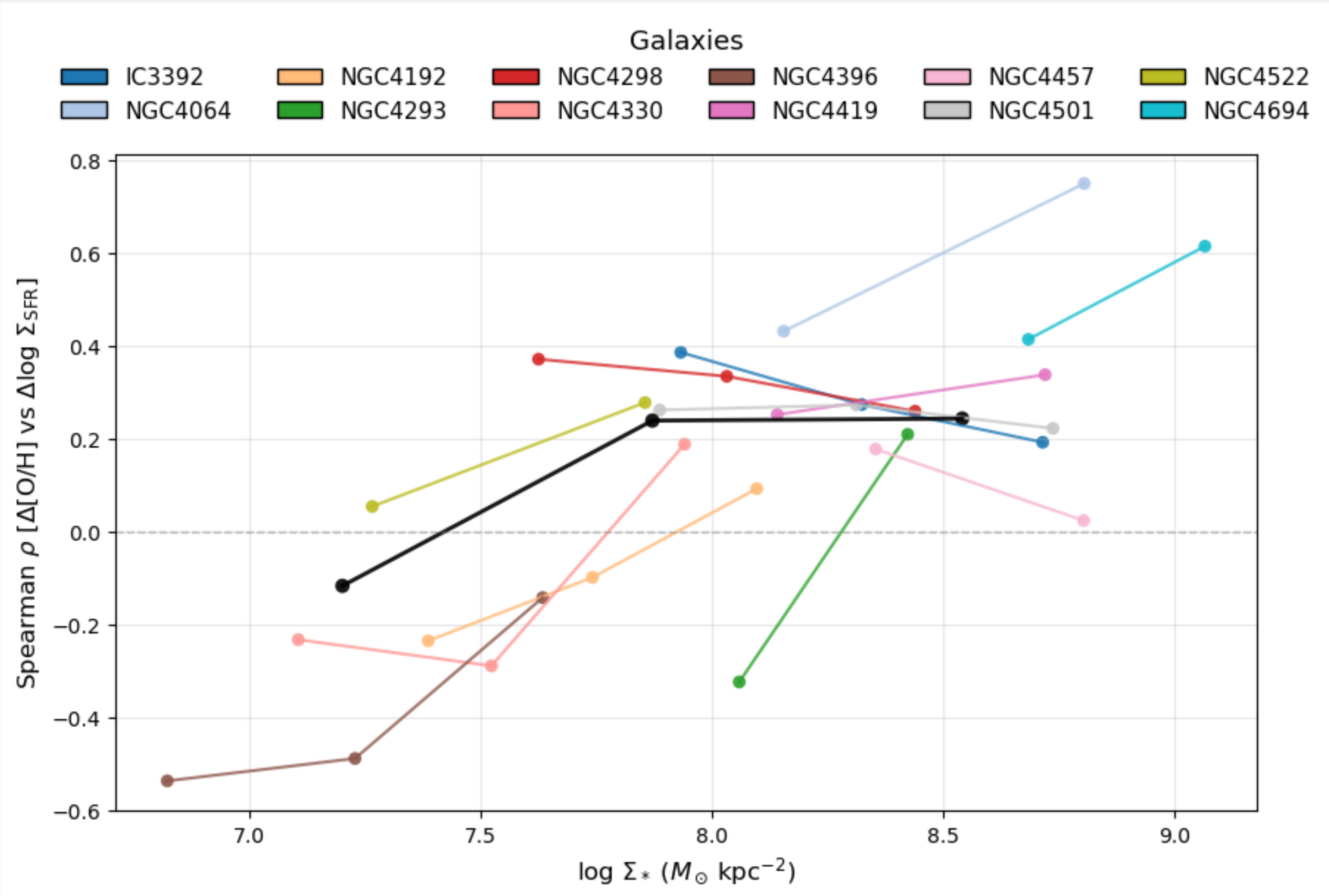
20251202 Redo Spearman Trend Binning Effect and a Closer Look at NGC4330

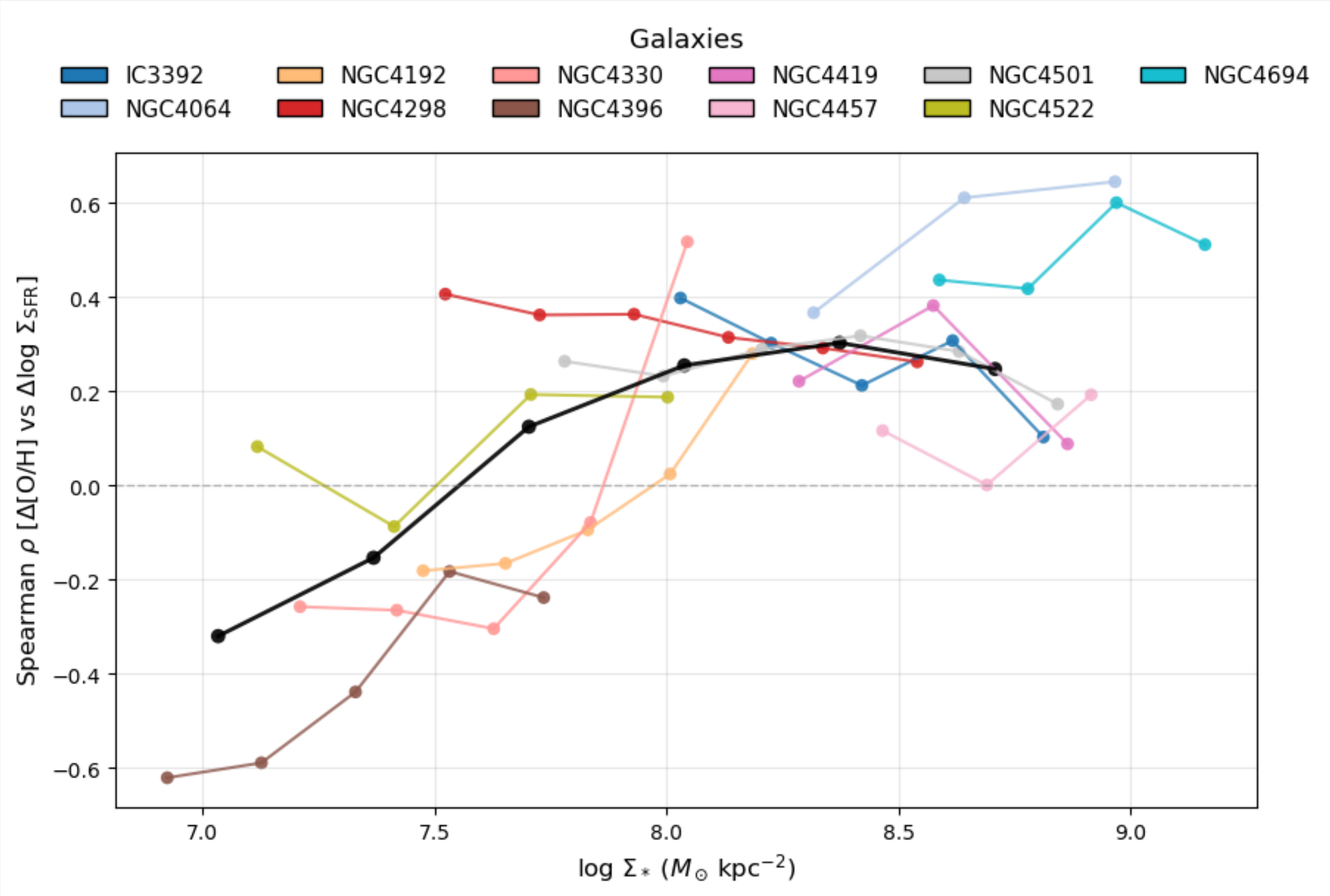
Now I have update the extinction correction in `Mass.py` to use the same Calzetti (2000) attenuation curve at `SFR+Z.py`. Interestingly, NGC4330 is no longer jumping up and down in Spearman trends.

$n = 3$



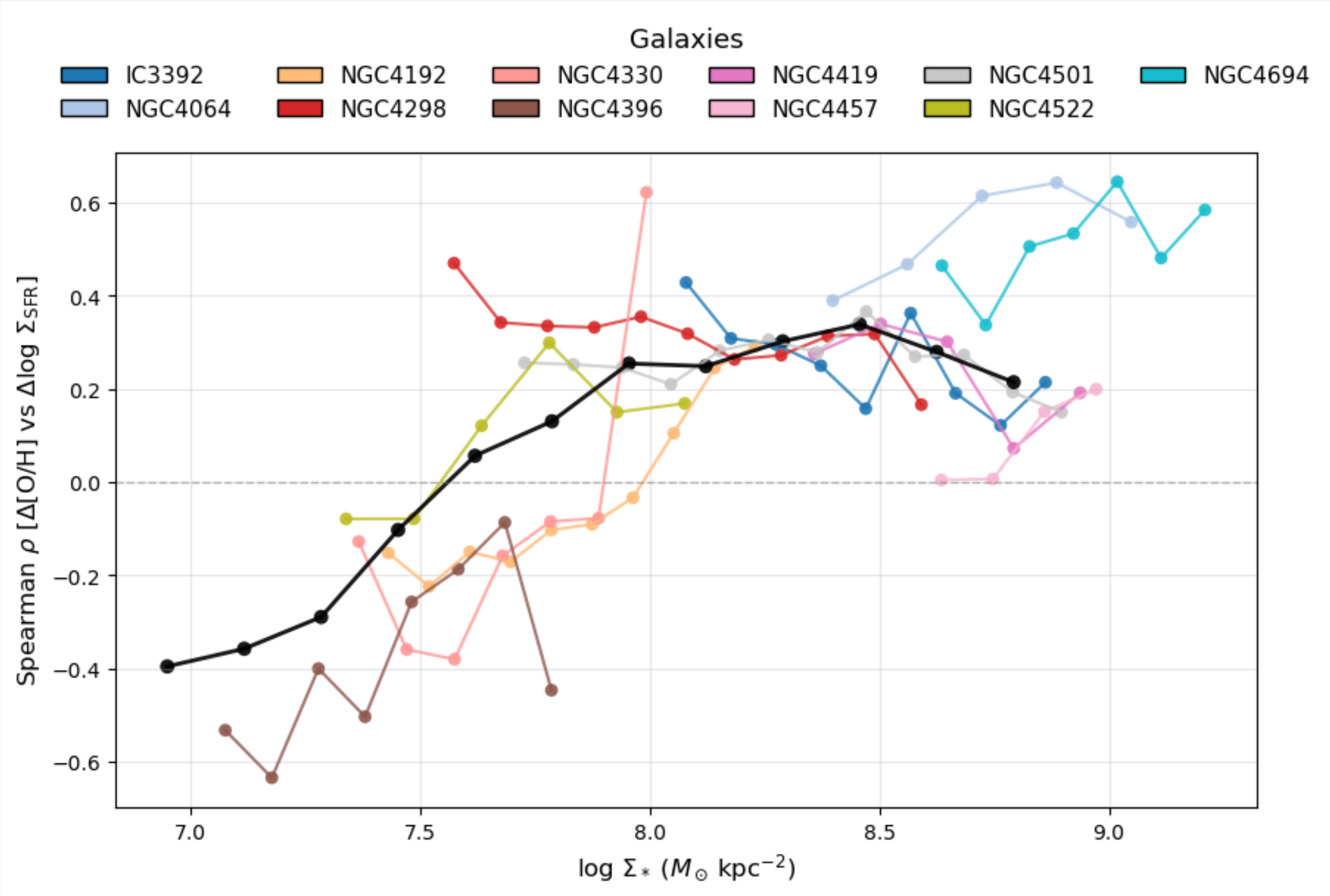
$n = 6$

But now seems NGC4522 is a bit problematic.



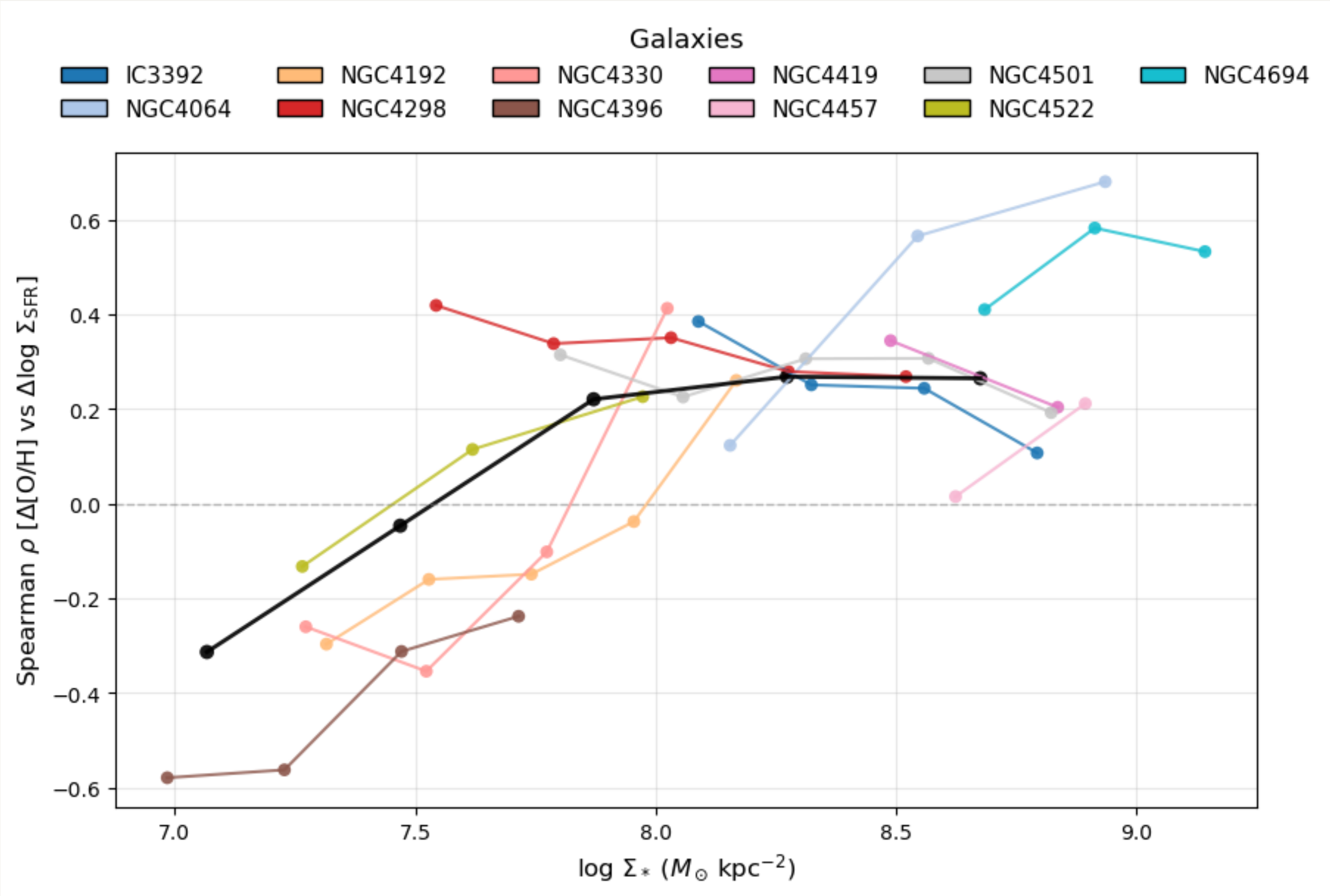
Galaxies
IC3392
NGC4064
NGC4192
NGC4298
NGC4330
NGC4396
NGC4419
NGC4457
NGC4501
NGC4522
NGC4694
Spearman ρ [Δ[O/H] vs Δlog Σ_SFR]
log Σ_* (M_⊙ kpc^−2)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
7.0
7.5
8.0
8.5
9.0

$n = 12$



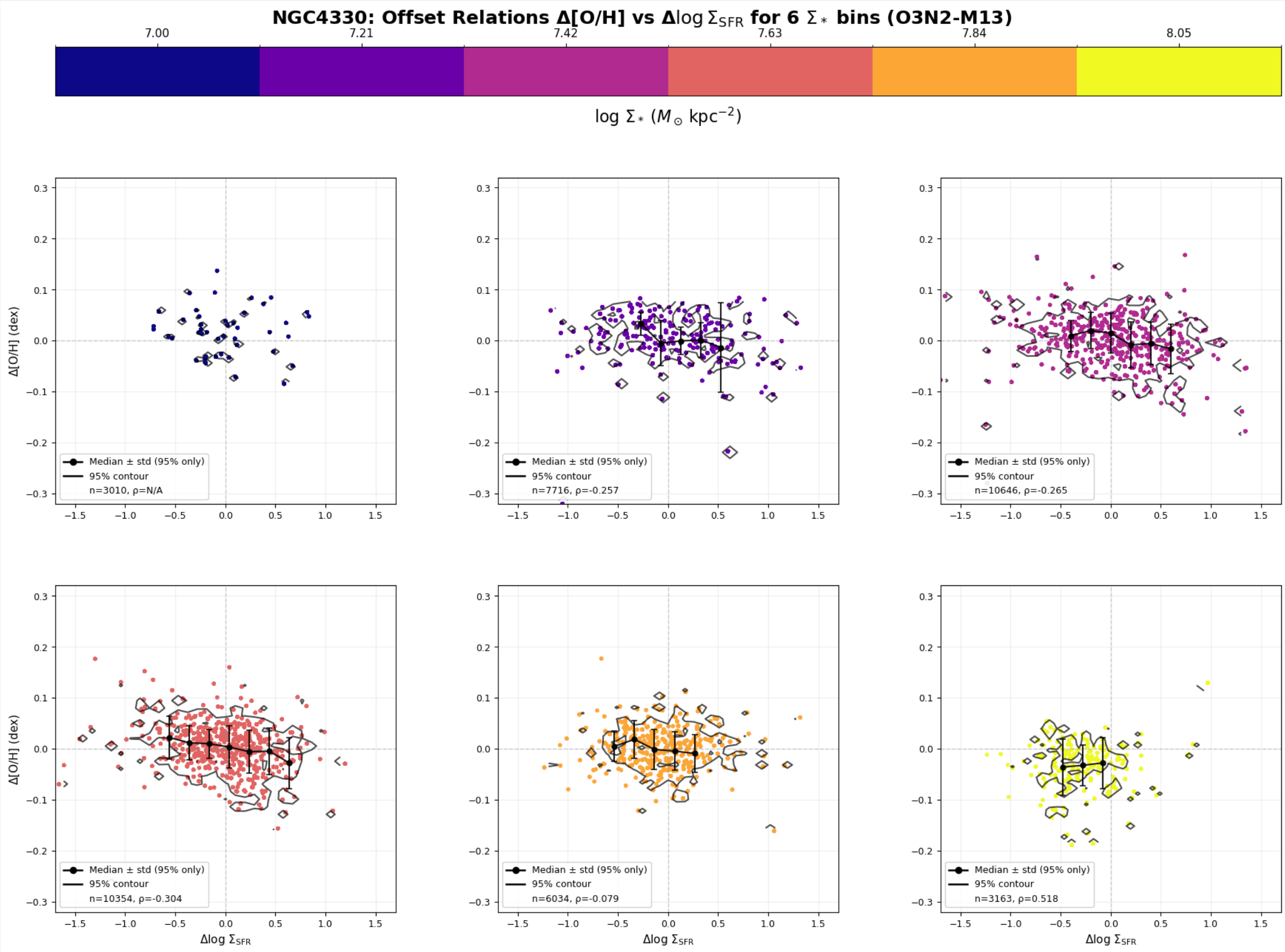
$n = 5$

So finally I think I find the sweet point at 5.



NGC4330

But actually if we look at NGC4330, we can still see that the highest mass bin is still biased by soem extreme points?


NGC4330: Offset Relations Δ[O/H] vs Δlog Σ_SFR for 6 Σ* bins (O3N2-M13)
7.00
7.21
7.42
7.63
7.84
8.05
log Σ* (M⊙ kpc⁻²)
Δ[O/H] (dex)
Δlog Σ_SFR
Median ± std (95% only)
95% contour
n=3010, ρ=N/A
n=7716, ρ=-0.257
n=10646, ρ=-0.265
n=10354, ρ=-0.304
n=6034, ρ=-0.079
n=3163, ρ=0.518